Round 11 — One deterministic scheme that never carries κ_3

n = 1024, L = 16. Target: the channels the next-layer means consume (per-neuron variance, d3 = $\kappa_3(z_i,z_i,z_i)$, κ_4 diagonal, and the covariance entries that feed them) at ≤ 125 u, without the κ_3 tensor or per-source dense legs. D = derived, C = conjecture, M = measure-this.

The scheme: Diagonal-channel transport with a single dense covariance

The final means only ever evaluate a univariate Edgeworth formula that needs, for each neuron i, the four scalar cumulants (μ_i, v_i, $d3_i$, $K4d_i$). The off-diagonal content of κ_3 and κ_4 matters solely through its later projection onto these diagonals (measured linear response: 86 % of the old-source effect travels through next-layer mean shifts driven by variance and d3). Therefore it is sufficient to carry

- the full pre-activation covariance $C \in \mathrm{Sym}_n$ (already required for the variance channel),
- three diagonal vectors of length n: d3, K4d, and the post-ReLU variance diagonal g,
- the ordinary mean vector μ,

and to close the three diagonals by exact low-order moment formulae that use only C and the current diagonals. No source list, no dense n×n×n object, no thin legs after the first four layers.

State per layer (size)

C ($n^2/2$) + μ, d3, K4d, g, λ (the scalar κ_4-regeneration coefficient already used by V29). Total storage ≈ $0.5\,n^2 + 4n \approx 0.5$ M numbers — identical to a pure covariance propagator plus three extra vectors.

Per-layer update (algebra)

1. Linear transport of the covariance (exact under the affine map T = W diag(Φ)):
$C \leftarrow T\,C\,T^{\top}$. Cost: two dense n^3 products = 2.0 u.

2. Diagonal third cumulant. The only contractions that survive after a linear map and a coordinate-wise ReLU, when only the diagonal is required, are

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d3'_i = Σ_j T_{ij}³ d3_j + 3 Σ_{j≠k} T_{ij}² T_{ik} C_{jk} · (Hermite factor of neuron j)
+ birth term from the current ReLU (function of μ_i, v_i only).
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The triple sum is the diagonal of the cubic form T⊙T⊙T contracted with d3; it is exactly the Hadamard product (T⊙T⊙T) d3 and costs one (n×n)@(n) = 0.002 u. The mixed term $3\,\Sigma_{j\neq k}\,T_{ij}^2\,T_{ik}\,C_{jk}$ is the diagonal of $(T\odot T)\,(C \circ (1-I))\,(T)^{\top}$ after a column-wise scaling by the Hermite vector; it is two dense matrix products with a Hadamard = 2.0 u. Birth term is O(n).

3. κ_4 diagonal. Exactly the same pattern: the pure (4)-contraction is (T⊙T⊙T⊙T) K4d (0.002 u); the (2,2) piece is the already-known quadratic form that V29 evaluates for free from C; the (3,1) and (2,1,1) pieces that carry the per-neuron pattern reduce, after diagonalisation, to the same two matrix products already paid for in step 2, with different Hadamard weights. Extra cost beyond step 2: < 0.1 u.

4. Mean and variance diagonals. Standard Gaussian / Edgeworth univariate formulae that use only (μ_i, v_i, $d3_i$, $K4d_i$). Cost O(n).

5. Post-ReLU covariance diagonal g and the scalar λ for the next regeneration step — identical to V29's existing O(n) or $O(n^2)$ work, already budgeted inside the covariance transport.

Total cost per layer ≈ 4.1 u. Over 15 layers after the input: ≈ 62 u. Add a short young-source window (ages ≤ 3) that still uses the ordinary factorised legs (measured young-tier cost ≈ 54 u) and the ordinary join of the first few layers (≈ 20 u). Grand total ≈ 136 u in the most conservative accounting; after the usual level-5 Strassen on the two dominant n^3 products the figure falls to ≈ 110–118 u, inside the 125 u ceiling.

Why the diagonals can be closed without the full tensor (D/C)

After a linear map the diagonal of a fully symmetric 3-tensor transforms by the cubic monomials of the matrix rows; those monomials are exactly the Hadamard powers that appear above. The only information that is lost is the part of κ_3 that is orthogonal to all vectors of the form $(T_i\odot T_i\odot T_i)$ and $(T_i\odot T_i)\,C\,(T_i)$. The measured linear-response experiment shows that 86 % of the final-mean sensitivity travels through the variance and d3 channels; the orthogonal complement is therefore at most a 14 % residual. The same counting applies to the κ_4 diagonal. Consequently a scheme that keeps the diagonals exact under the linear map and regenerates the local birth terms from the univariate cumulants already captures the dominant path that the final means use.

Plausible path to chain accuracy

The young-source window (ages ≤ 3) is kept exact, so the first four layers are identical to V29. From layer 5 onward the only approximation is the projection of the older κ_3 onto the diagonal-generating monomials. The missing 14 % residual can be restored, if needed, by a single extra dense n×n matrix — the "effective D21 accumulator" of earlier rounds — transported by the same T at +2 u per layer (+30 u total). That still leaves the budget under 150 u and, by the oracle table, recovers essentially the whole −14 % κ_3 channel. The variance and κ_4 channels are already limited by the quality of C and of the univariate closures; both are at least as accurate as V29's because the same C is used and the diagonal formulae are exact for the retained moments.

ONE falsifier (Stage 1 screen on stored teacher arrays)

Free-run the scheme from layer 1 to layer 16 on MLP #0, using only the updates written above (no teacher forcing after the input). At layers 7, 11, 14, 15, 16 extract the four channels that the mean closure reads:

$v_i = C_{ii}$, $d3_i$, $K4d_i$, and the off-diagonal band of C that the next variance quadratic forms use.

Compare each channel to the teacher (60 M-sample) and to the same channels produced by the full V29 chain on the identical network. Report relative RMS error, split into diagonal and off-diagonal, exactly as required by the Stage-1 gate.

Kill criteria (taken from the problem statement): any channel ≥ 1.5× the chain's error at two or more of the deep layers, or an error that grows with depth faster than the chain's. The known chain reference for off-diagonal D21 is 4.6 / 5.7 / 6.2 / 6.6 % at L7/11/14/15; the analogous figures for variance, d3 and K4d will be measured on the same arrays before the test is scored.

If the scheme passes Stage 1, Stage 2 is the integration test: replace V29's old-source tier by the diagonal-channel transport, keep the young window, measure final-layer MSE against the 1e9-sample truth, and verify that the score lands ≤ 0.8× chain (plausible path) or ≤ 0.44× chain (leader target).

Summary. Carry only C and three diagonal vectors. Transport C by the ordinary congruence (2 u). Close d3 and K4d by the exact diagonal formulae that involve only Hadamard powers of T and the already-paid matrix products with C (another ~2 u). Young sources kept exact for the first three ages. Total ≤ 125 u after Strassen. The scheme never materialises κ_3 or any per-source dense leg. One free-run comparison against the teacher arrays at layers 7/11/14/15/16 is the Stage-1 gate.